

AI & IT Operations Assistant PoC

Project Documentation - Internal IT & Process Self-Service Assistant

Isolated Microsoft 365 tenant | Fictional policies and test data | Prototype only

Project type	Proof of Concept	Status	Core components tested
Stack	Copilot Studio, Power Automate, SharePoint, Outlook	Production use	No

1. Executive summary

The PoC addresses a common internal-operations problem: employees ask repetitive IT questions and submit access requests through unstructured messages. This creates incomplete requests, manual follow-up, limited status transparency, and inconsistent approval records.

The solution combines a grounded Copilot Studio assistant with deterministic Power Automate workflows. The AI layer handles natural-language questions and extracts structured request information. The workflow layer handles predictable business actions: status changes, approval routing, record updates, and notifications.

2. Business process and target state

- Repetitive IT/process questions consume employee and IT time.
- Access requests may be vague, incomplete, or sent through inconsistent channels.
- Approvals are difficult to track when they happen through scattered messages.
- Employees may request permissions broader than necessary.
- An ungrounded AI assistant may invent internal policy details.

Target process:

Employee	Copilot	SharePoint	Power Automate	Manager	Status	Notification
-----------------	----------------	-------------------	-----------------------	----------------	---------------	---------------------

3. Design decision: AI, deterministic automation, or both?

Area	Technology choice	Reason
Natural-language questions	AI / Copilot Studio	Users express the same need in many different ways.
Knowledge grounding	AI + controlled documents	Use approved sources and avoid inventing internal policy.
Request capture	AI extracts structured fields	Turn conversation into system, justification, priority, approver, and description.
Approval and status logic	Deterministic Power Automate	Business actions must be predictable, auditable, and repeatable.
Access decision	Human in the loop	Sensitive permissions should not be granted autonomously by AI.

4. Implemented components

Microsoft 365 / Power Platform setup: Tenant, user, licenses, Power Platform environment, Dataverse, and security roles.

SharePoint tracking: IT Requests list as the structured system of record.

Approval workflow: New request -> Pending Approval -> approval -> Approved/Rejected -> update -> email.

Copilot Studio assistant: Grounded internal IT/process assistant using three fictional policy documents.

Safety behaviour: Least privilege, no credentials, escalation for unknown information, no automatic access granting.

Request-creation tool: Agent tool collects six fields and is designed to create a SharePoint request.

5. Test plan and results

Test case	Expected	Actual	Status
Valid access request	Approval completes	Approved + email	Passed
Excessive Global Admin request	Reject using least privilege	Rejected + comments	Passed
Exact policy retrieval	Return 2-business-day target	Refined and retested	Passed after iteration
Unknown internal question	Escalate, do not invent	Escalated	Passed
Sensitive data / credentials	Refuse unsafe handling	Refused/warned	Passed
Tool input extraction	Populate six request fields	All six extracted	Passed
Agent -> workflow invocation	Create SharePoint item	HTTP 403 before run	Known limitation

6. Known limitation and diagnosis

The Copilot agent correctly identified the request-creation intent and populated all six workflow inputs. The tool call returned HTTP 403. Monitoring showed zero workflow runs, indicating that the request was blocked at the agent-to-workflow authorization boundary before SharePoint or workflow logic executed.

Planned troubleshooting:

- Validate runtime entitlement and licensing in a fully licensed tenant.
- Validate environment ownership and runtime permissions.
- Test the bridge using the classic Agent Flow experience if needed.
- Use a dedicated least-privilege service identity when appropriate.

7. Security and governance considerations

- Least privilege: broad administrator access is challenged; minimum permissions are preferred.
- Human approval: the AI does not grant access autonomously.
- Prompt safety: passwords, MFA codes, API keys, tokens, and private keys must not be requested or stored.
- Data minimization: only information required for the process is collected.
- Grounding and escalation: unsupported internal questions are escalated instead of answered from general knowledge.
- Production separation: real deployment should use controlled Dev/Test/Prod environments, DLP policies, monitoring, and formal ownership.

8. Production roadmap

Resolve runtime integration: Fix agent-to-workflow authorization in a fully licensed environment.

Improve knowledge architecture: Move demo files to permission-aware SharePoint sources where appropriate.

Add status self-service: Allow employees to retrieve their own request status.

Add resilience: Error handling, retries, owner notifications, logging, and monitoring.

Pilot with a small group: Validate usefulness, answer quality, failure modes, and support needs.

Measure value: Self-service rate, approval time, flow errors, escalation rate, and user satisfaction.

Productionize safely: Permission model, DLP, GDPR review, support ownership, documentation, and rollback plan.

9. Evidence available

- Compact interview presentation with screenshots
- Approval and rejection test runs
- SharePoint request tracking list
- Approval and notification emails
- Grounded-answer and least-privilege tests
- Tool input extraction and HTTP 403 evidence
- Three fictional policy documents used by the agent

10. Conclusion

This is a PoC, not a production deployment. It demonstrates the implementation mindset required for the role: understand a process, identify automation potential, choose AI and deterministic automation deliberately, build a prototype, test positive and negative paths, document limitations, and define a safe path toward real operations.